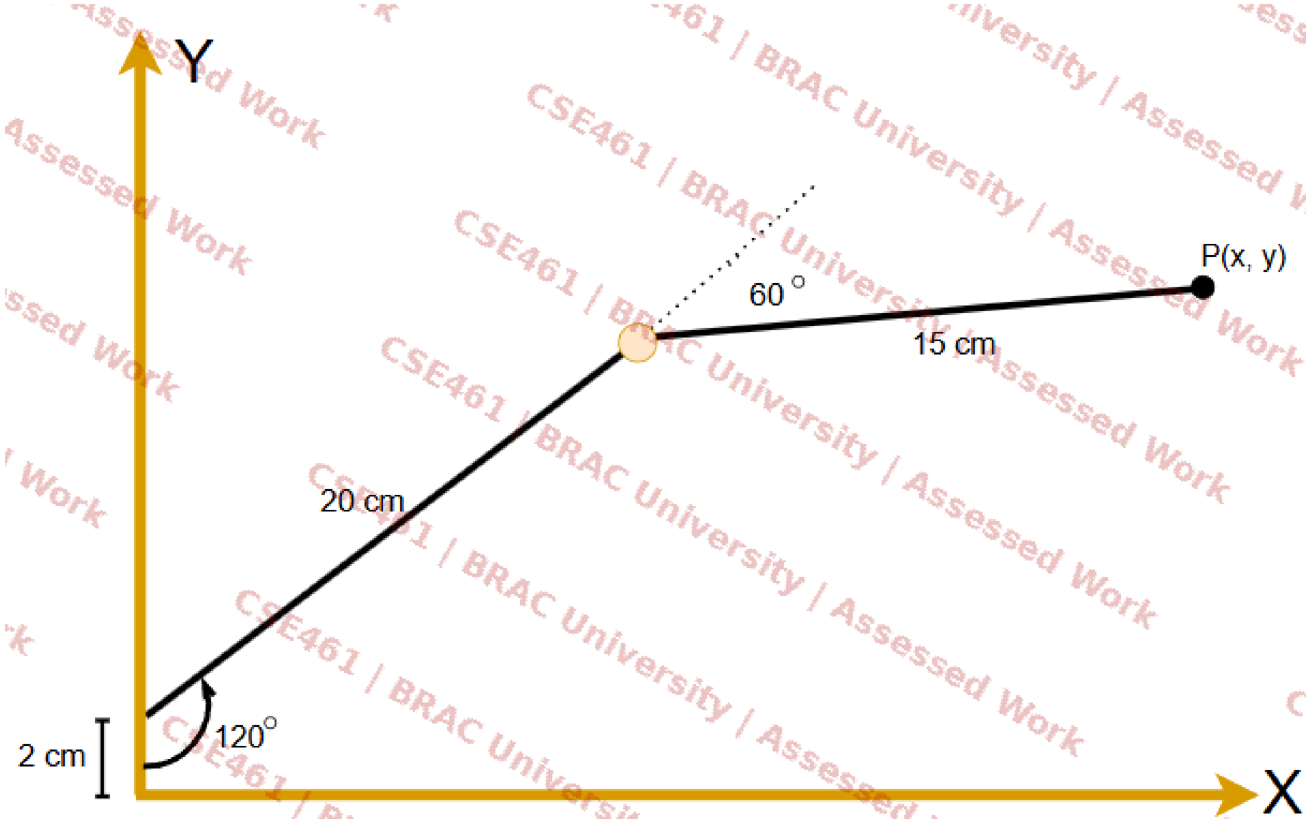
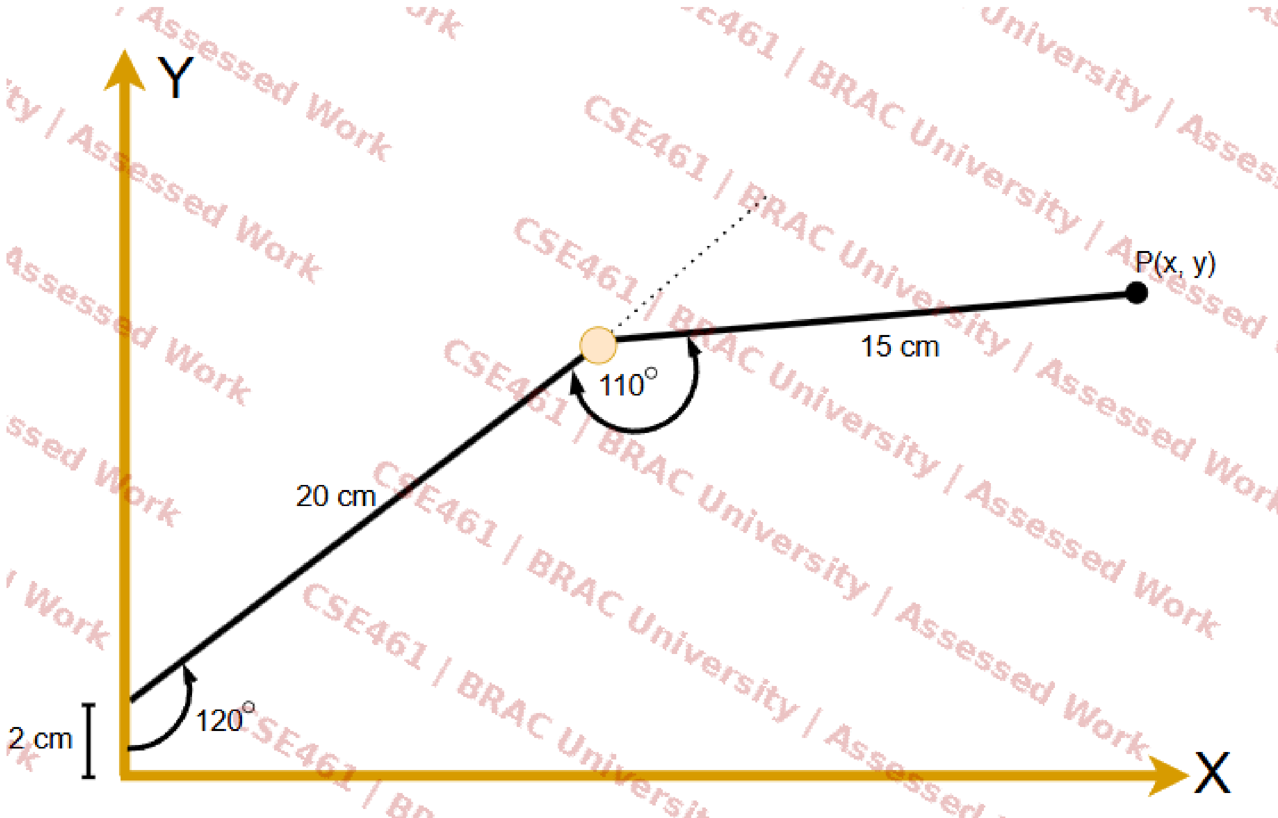
Question 1:



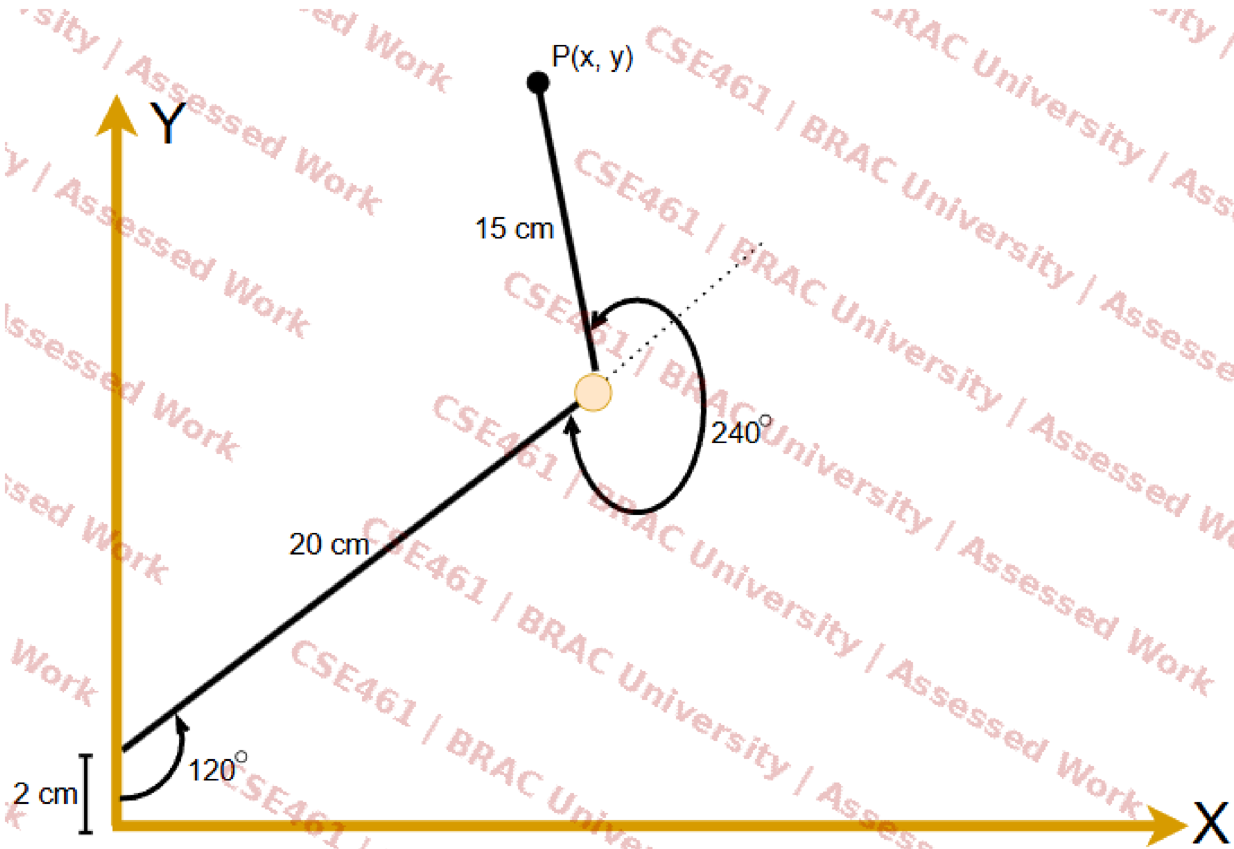
What is the value of point P? [Use both trigonometry and DH method both, the answer will be identical]

Question 2:



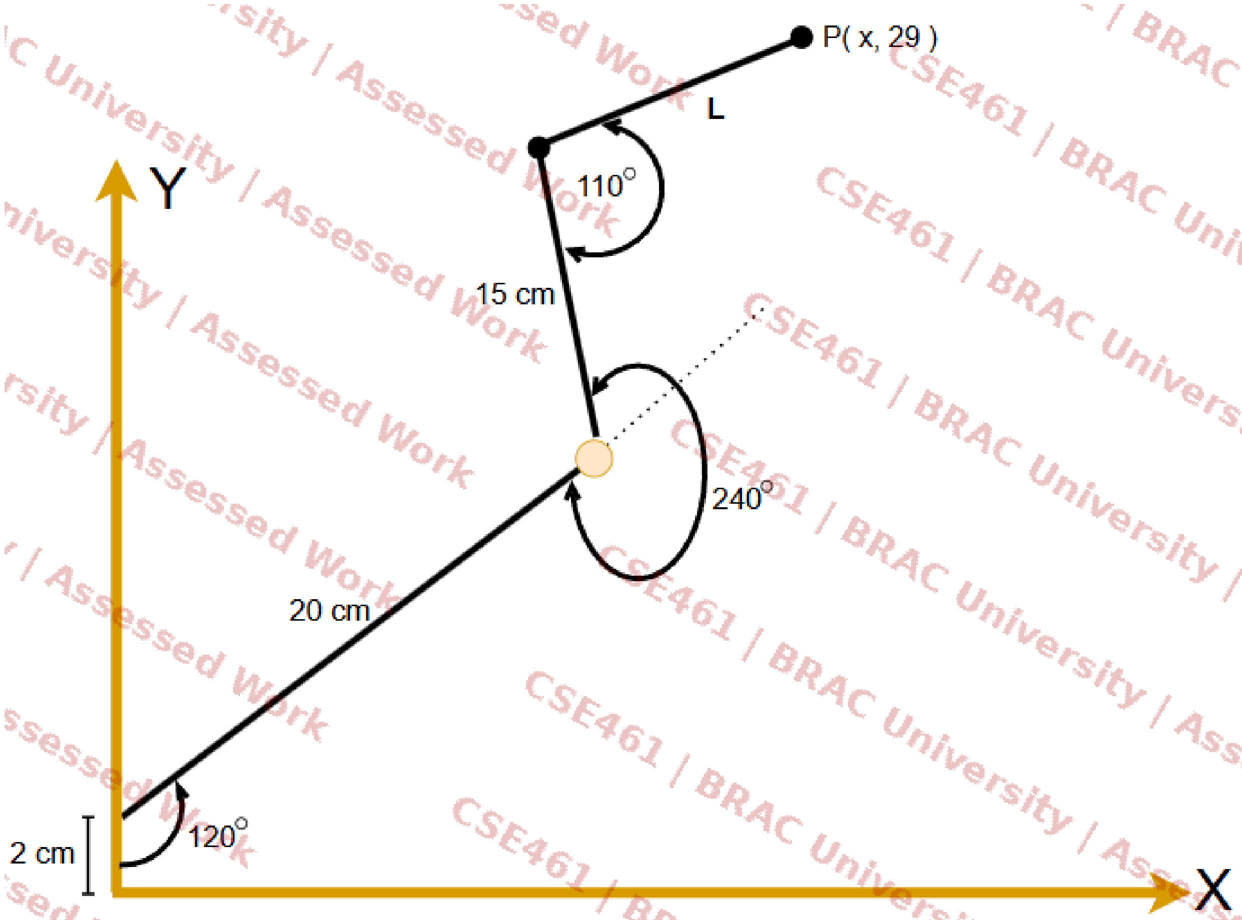
What is the value of point P? [Use both trigonometry and DH method both, the answer will be identical]

Question 3:



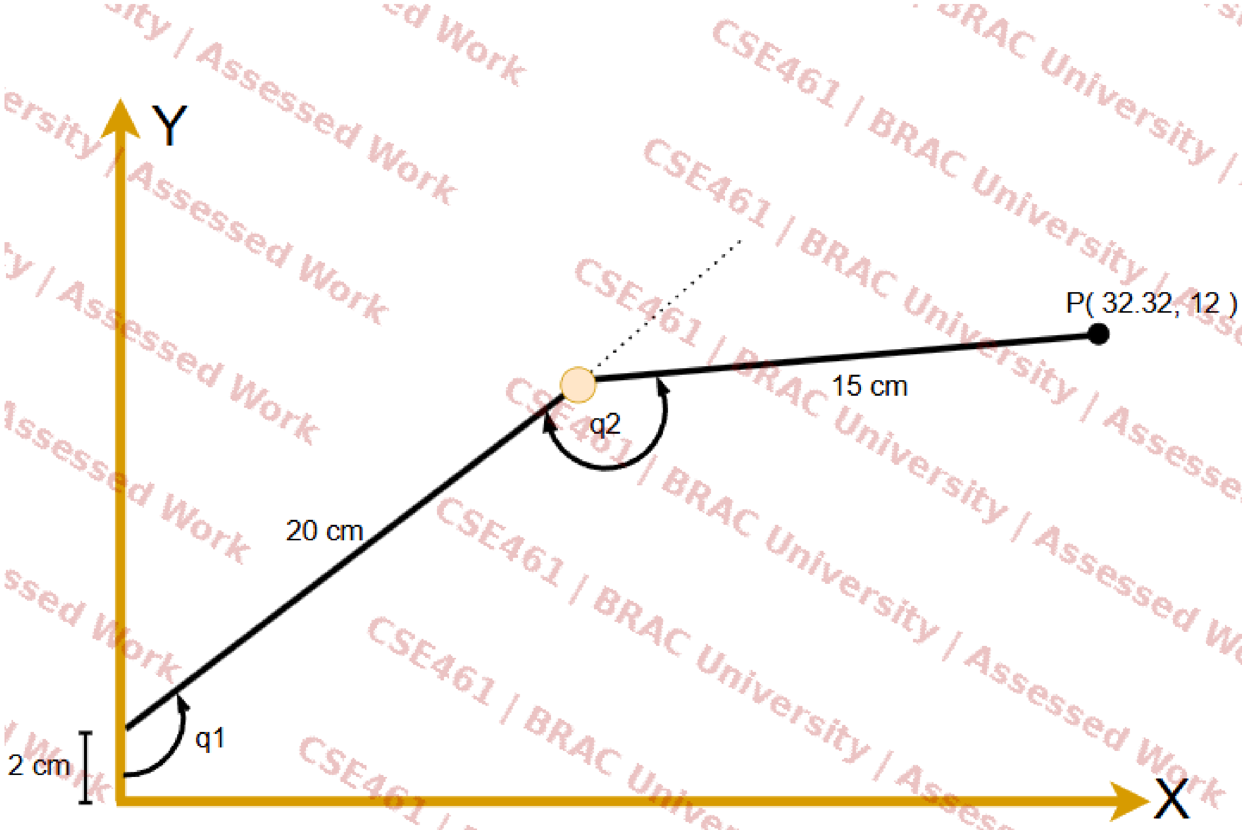
What is the value of point P? [Use both trigonometry and DH method both, the answer will be identical]

Question 4:



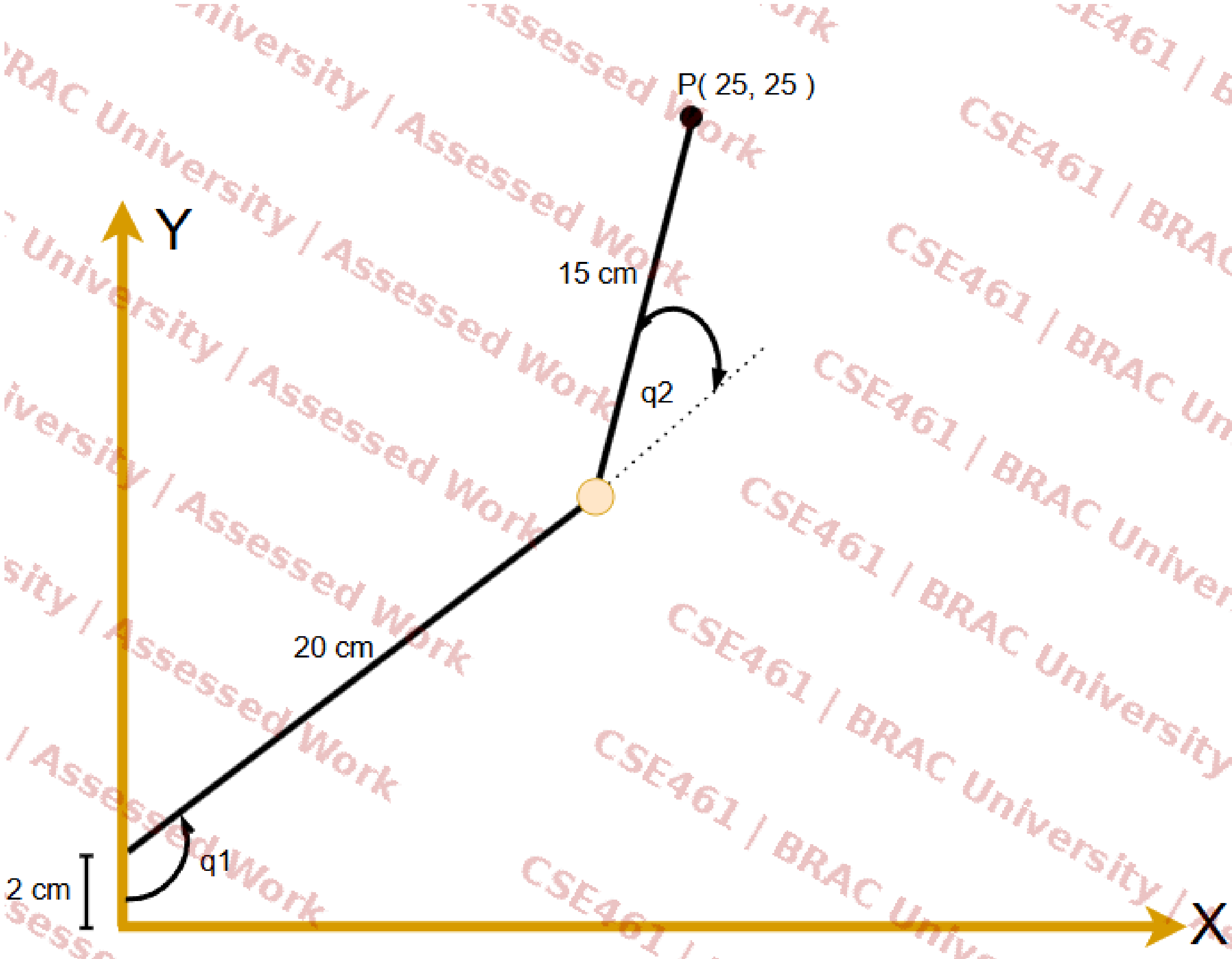
What is the value of L and x ?

Question 5:



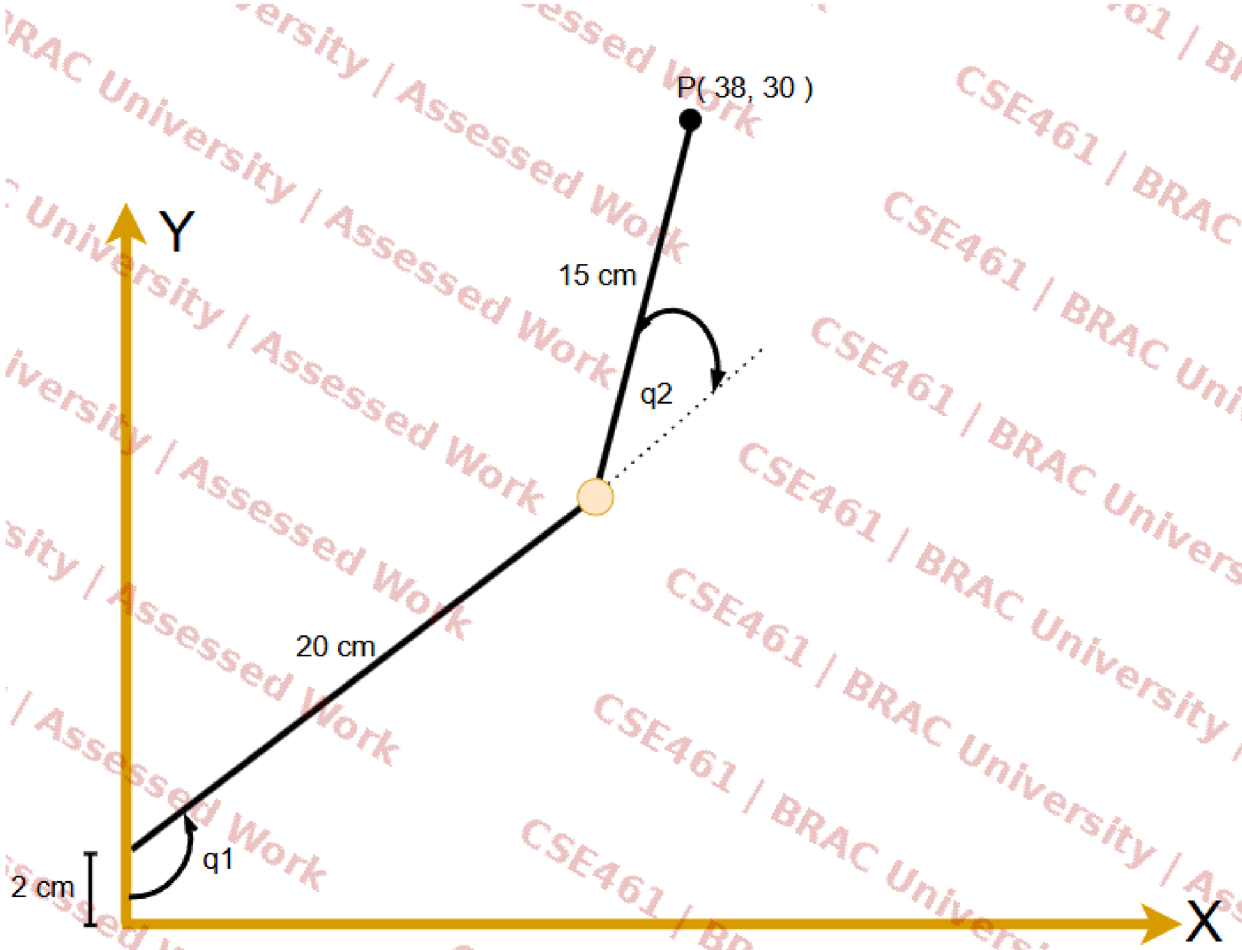
What is the value of q1 and q2?

Question 6:



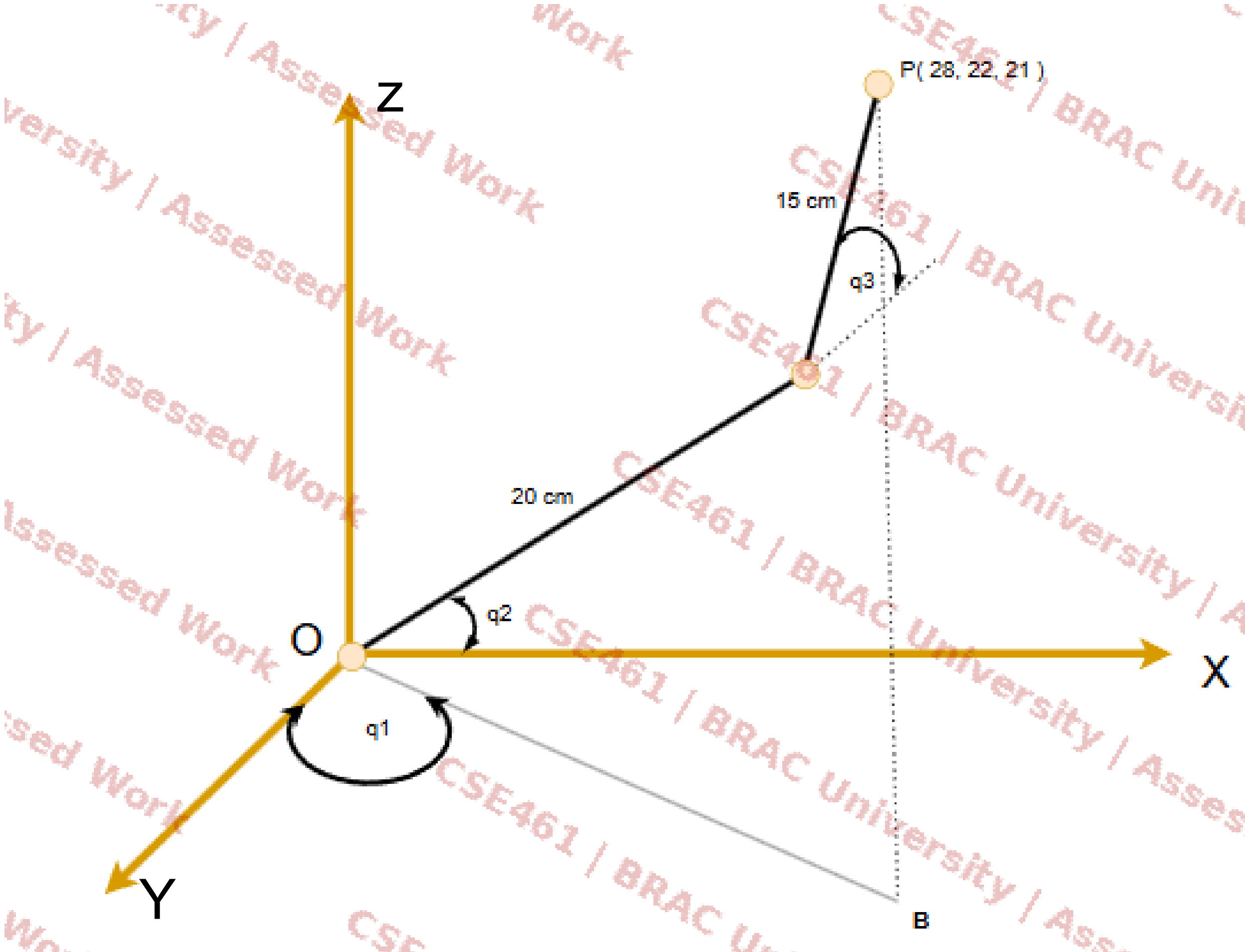
What is the value of q1 and q2?

Question 7:



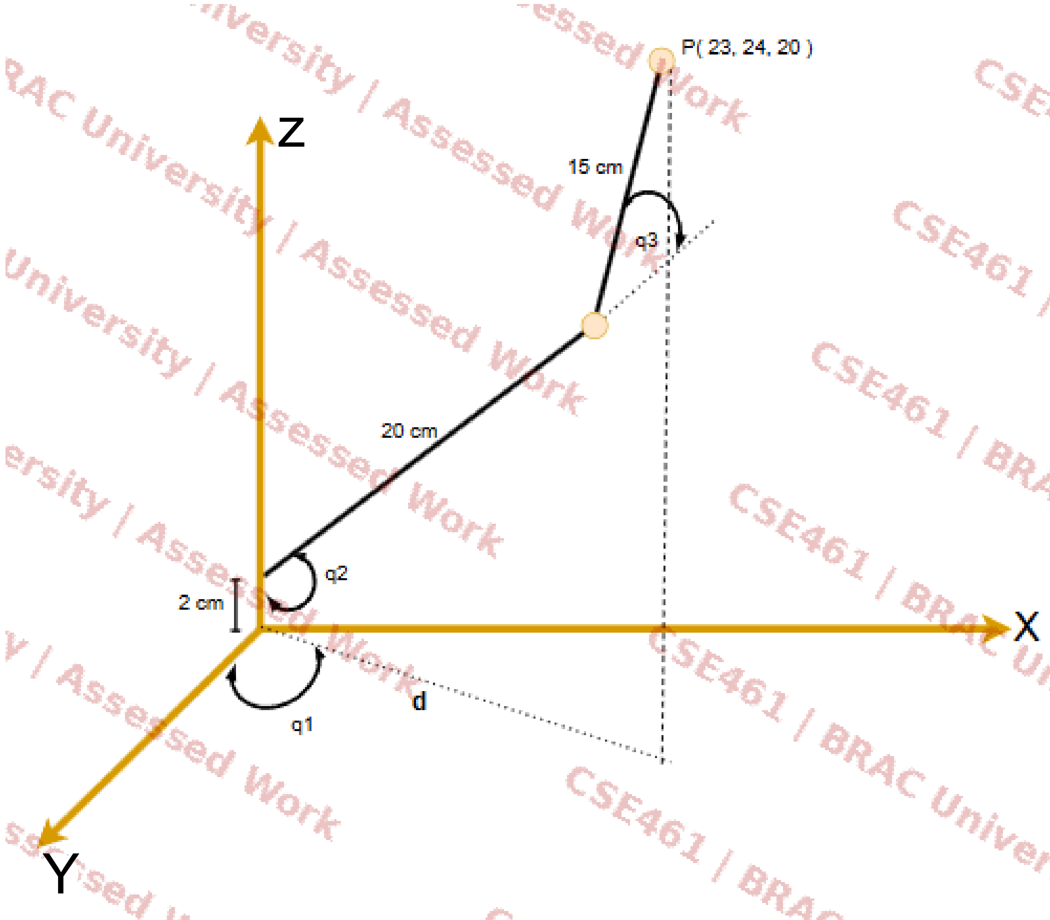
What is the value of q1 and q2 ?

Question 8:



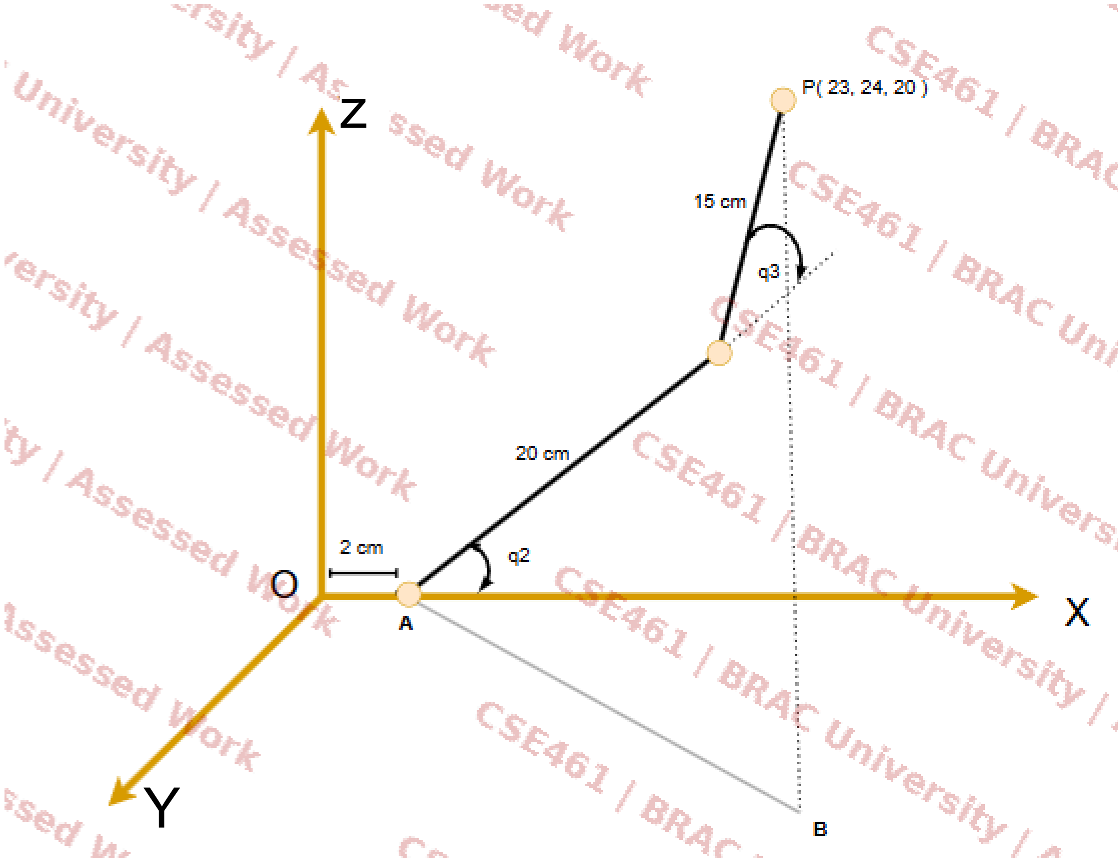
What are the values of q1, q2 and q3?

Question 9:



(a) What does d indicate in the diagram.
(b) Determine the value of d?
(c) What are the values of q1, q2 and q3?

Question 10:



(a) What is the value of AB?
(b) What are the values of`q2 and q3?
(c) What is the angle between Z axis and AB?

Question 11:

Given DH parameters

Joint (i)	θ (deg)	α (deg)	a (m)	d (m)
1	−30°	0°	0.80 m	0
2	60°	90°	0.50 m	0
3	−45°	0°	0.60 m	0.40 m

DH Convention: Denavit-Hartenberg. All revolute joints rotate about the Z-axis of the current frame.

Using the DH parameters provided in the table above, perform the following tasks:

(a) Draw the robot configuration clearly:
- Top View (X–Y plane): Show the projection of all links and joints. Indicate the joint angles (θ) with proper reference directions.
- Side View (X–Z plane): Show the vertical structure of the robot, including link lengths (a) and offsets (d). Label all joints and coordinate axes (X, Y, Z).

(b) Using a trigonometric (geometric) approach, determine the final position (x, y, z) of the robot's end-effector. Show all necessary steps clearly.